Syntactic Complexity and Model Robustness

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1 Introduction

This report investigates the impact of syntactic complexity on the robustness of NLP models using the MultiNLI dataset. To evaluate this, I implemented three complexity metrics—Clause Count, Subject-Verb Distance, and CFG Tree Depth—and developed an automated pipeline to apply meaning-preserving perturbations, specifically passive voice transformations, adjective-to-relative-clause rewrites, and prepositional phrase insertions. I compared the performance of four architectures: a simple MLP, an RNN, and two Transformer-based models, RoBERTa and BART. My findings reveal that while Transformers achieve superior baseline accuracy, they are vulnerable to structural shifts such as SVO to OVS. Error analysis indicates that these performance regressions are driven by difficulty resolving inverted predicate-argument structures and long-distance dependencies, with Transformers exhibiting increased sensitivity when processing out-of-domain text.

2 Complexity Metrics

I measured syntactic complexity using three metrics, ranked from simplest to most complex. CFG tree depth captures how deeply nested a sentence's phrase structure is. Subject-verb distance captures how tangled a specific dependency relationship is, and clause count captures how much is happening in a sentence overall. My understanding is that together, the three metrics cover different dimensions of syntactic complexity rather than redundantly measuring the same thing.

1. **Clause Count** — I define clause count as the number of verbs and auxiliaries in a sentence. Each verb roughly corresponds to a clause, so more verbs indicate more predicate-argument structures. For example, *"The cat ate fish"* has a clause count of 1 (verb = *ate*), while *"The student whose parents died is back to write the exam that is difficult"* has a count of 4 (verb = *died*, *write*; aux = *is back*, *is difficult*). Initially, I only used verbs, but this yielded a very low clause count for complex sentences. Auxiliaries were therefore included as well. **Rationale:** Valizadeh and Soltanpour (2021) argue that syntactic complexity develops in stages, with more clauses emerging at higher proficiency levels, directly supporting clause count as a valid complexity indicator. Moreover, managing multiple clauses places higher attentional demands on the processor — whether human or model — providing a measure of how syntactically complex the model perceives the text (Kellogg, 1990).

2. **Subject-Verb Distance** — This is the number of words separating the main verb of a clause from its subject noun. For example, in *"The student who studied all night passed the exam,"* the subject *student* and verb *passed* are separated by 5 tokens. **Rationale:** Longer distances indicate more intervening material to be parsed before the core subject-predicate relationship is resolved, making this a direct indicator of syntactic complexity.

3. **CFG Tree Depth** — This is the maximum depth of the constituency parse tree of a sentence. For example, *"The cat ate fish"* has a CFG depth of 4, while *"The student whose parents died is back to write the exam that is difficult"* has a depth of 11, reflecting its two nested relative clauses. **Rationale:** Deeper trees indicate more nested structures, which require resolving more hierarchical relationships simultaneously, providing a strong indication of syntactic complexity.

3 Model Architecture & Baseline Accuracy

I used the following model architectures on the unperturbed MultiNLI subset of 15000 samples.

1. **MLP** - A lightweight feedforward network trained on frozen RoBERTa-base embeddings. The MLP consists of a single hidden layer (256 units) with ReLU activation and dropout. Its primary advantage is simplicity and fast inference; however, it cannot capture sequential

or hierarchical structure in the input, making it the weakest architecture for NLI.

2. **RNN** - A single-layer LSTM trained on the same frozen RoBERTa-base embeddings. The LSTM can capture sequential dependencies across the input, giving it a structural advantage over the MLP. However, it still relies on frozen embeddings and lacks the contextual depth of full Transformer models.

3. **RoBERTa-large-mnli (Encoder-only Transformer)** - A large pretrained Transformer encoder fine-tuned directly on MultiNLI. Such models process the entire input bidirectionally, making them well-suited for classification tasks. This is the strongest architecture for the task.

4. **BART-large-mnli (Encoder-Decoder Transformer)** - A sequence-to-sequence Transformer fine-tuned on MultiNLI. BART's encoder produces rich contextual representations suitable for classification. Its advantage over encoder-only models is flexibility; however, it introduces more parameters and complexity than are needed for classification.

Model	Baseline Accuracy
MLP	0.5572
RNN	0.5873
RoBERTa-large-mnli	0.9019
BART-large-mnli	0.8998

Table 1: Baseline accuracy of each model on the unperturbed MultiNLI subset.

4 Perturbation Methodology

4.1 Approach

The goal of perturbations is to increase the syntactic complexity of the premise while preserving the original meaning and ground-truth NLI label. All perturbations are applied to the premise only, and each perturbation is applied separately, producing three distinct perturbed datasets in addition to the original.

1. **Adjective-to-Relative-Clause (adjrel)** - This perturbation rewrites adjectival modifiers as relative clauses, increasing syntactic complexity by introducing subordinate clause structures. I identify tokens with the `amod` dependency relation and rewrite them as relative clauses attached to their head noun. The relative pronoun is selected based on animacy (*who* for persons, *which* for objects). Hyphenated compounds (e.g., *long-term*) are skipped to avoid malformed output. Example from `data.jsonl`:

   > *Original:* "The Herron School of Art and Gallery at IUPUI is contemporary art!"
   > *Perturbed:* "The Herron School of Art and Gallery at IUPUI is art, which is contemporary!"

2. **Passive Voice Transformation (passive)** - This perturbation converts active to passive voice, disrupting the subject-verb-object word order. I identify sentences with a clear subject (`nsubj`) and direct object (`dobj`), then reconstruct the sentence in passive form using `pyinflect` to generate the correct past participle. If no valid subject-object pair is found, the sentence is returned unchanged. Example from `data.jsonl`:

   > *Original:* "Oh, it can lead one to a rhymed."
   > *Perturbed:* "One was led by it to a rhymed."

3. **Prepositional Phrase Insertion (pp).** This perturbation inserts a very generic prepositional phrase after the first noun phrase of the premise, increasing subject-verb distance without altering meaning. Location-based phrases (e.g., *in the room*) were deliberately avoided as they could introduce new information that contradicts a hypothesis. Instead, I use discourse-level and temporal phrases which carry no referential content. Sentences where the first noun phrase ends with a contraction are skipped to avoid malformed output. The full list of phrases used is provided in Appendix A. Example from `data.jsonl`:

   > *Original:* "Credibility is a vital factor, and Jim Lehrer does, indeed, have it."
   > *Perturbed:* "Credibility at that point is a vital factor, and Jim Lehrer does, indeed, have it."

4.2 Complexity Metric Comparison

Table 2 shows the average complexity scores before and after each perturbation. All three perturbations

show and increase in at least one complexity metric relative to the original, confirming that the perturbed sentences are syntactically more complex.

Perturbation	Clause Count	S-V Dist	CFG Depth
Original	3.29	2.34	9.25
adjrel	4.34	2.99	10.67
passive	3.11	3.12	8.71
pp	3.55	4.29	9.62

Table 2: Average complexity scores before and after each perturbation.

5 Model Performance

Table 3 summarizes the accuracy of each model on the original and perturbed datasets.

Model	Orig.	adjrel	passive	pp
MLP	0.5572	0.5479	0.5438	0.5486
RNN	0.5873	0.5733	0.5671	0.5774
RoBERTa	0.9019	0.8973	0.8287	0.9004
BART	0.8999	0.8967	0.8271	0.8989

Table 3: Accuracy of each model on original and perturbed datasets.

Transformers outperform simple models. RoBERTa and BART achieve approximately 90% accuracy on the original dataset, compared to 55–58% for the MLP and RNN. This is because they exhibit bidirectional attention, which capture deep semantic relationships between premise and hypothesis that frozen-embedding models cannot.

Passive voice is the most disruptive perturbation. The passive voice perturbation causes the largest accuracy drop across all models, with RoBERTa and BART both dropping approximately 7.3% (from 90% to 83%). Transformer models trained on NLI data develop strong positional rules — expecting the subject to precede the verb and the object to follow it. Passive voice directly violates this word order, disrupting the model's ability to correctly identify predicate-argument relationships.

Simple models are paradoxically more robust to passive voice. The MLP and RNN only drop approximately 1.3–2% on the passive perturbation, despite being weaker overall. This is because the embedding itself does not encode positional structure in the same way that full attention-based models do.

Adjective-to-relative-clause perturbation has minimal impact. All models drop less than 1% on the adjrel perturbation. This is likely because relative clause constructions are common in natural language and well-represented in the MultiNLI training data, making them familiar to all model architectures.

Prepositional phrase insertion is nearly invisible to models. RoBERTa and BART drop less than 0.2% on the pp perturbation, suggesting they correctly learn to ignore semantically generic modifiers. The MLP and RNN show a slightly larger drop ($\sim$0.9%), consistent with their greater sensitivity to increases in subject-verb distance.

6 Error Analysis

I identify cases where each model correctly classified the original premise-hypothesis pair but failed on the perturbed version. These failures are then analyzed across perturbation types and data splits. The error summary on a random sample of 2,000 data points can be found in Appendix B.

6.1 Adjective-to-Relative-Clause Failures

The RNN shows higher failure rates on adjrel (7.61%) compared to transformers (2.40% for RoBERTa). The following example illustrates why:

> *Hypothesis:* "The Praya is situated behind the ferry pier."
> *Original premise:* "The Praya, the promenade in front of the ferry pier, is a good place to observe the many junks and fishing boats in the harbor."
> *Perturbed premise:* "The Praya, the promenade in front of the ferry pier, is a place, which is good, to observe the junks, which are many, and fishing boats in the harbor."

The added relative clauses introduce multiple long-distance dependencies that the RNN's single CLS embedding cannot fully resolve. Transformers handle this better due to their bidirectional attention, which can parse relevant tokens regardless of distance.

6.2 Passive Voice Failures

Passive voice causes the highest failure rates for both transformer models, with RoBERTa failing on 9.66% and BART on 9.37% of originally-correct examples. The following example illustrates a typical failure:

Hypothesis: "The H-2A worker is being forced to leave the country."
Original premise: "The H-2A worker must depart the country and is subject to deportation for failing to do so."
Perturbed premise: "The country was departed by the H-2A worker."

The original premise entails the hypothesis (label 0), and RoBERTa correctly identifies this. However, after passive transformation, RoBERTa fails. This is because the model expects the semantic agent to appear before the verb. Passive voice inverts this, placing the patient ("the country") in subject position, disrupting the model's ability to resolve the predicate-argument structure correctly. Additionally, some passive transformations produced malformed output (e.g., "Her shack was entered by I"), which further degrades model performance.

6.3 Prepositional Phrase Insertion Failures

Although prepositional phrase insertion causes the lowest overall failure rates (1.17% for RoBERTa, 1.33% for BART), the failures that do occur are largely caused by malformed perturbations rather than genuine syntactic complexity. Inspection reveals that the insertion pipeline occasionally produces ungrammatical output when applied to short or contracted tokens, for example:

Original premise: "that's right you can work yourself to death"
Perturbed premise: "thatwhich we are talking about 's right you can work yourself to death"

In such cases, the model correctly classified the original but failed on the malformed version, not because of increased syntactic complexity, but because the perturbation introduced noise. This highlights a limitation of rule-based perturbation pipelines: they can produce ungrammatical output on informal or spoken text, which is present in the MultiNLI matched split (telephone speech genre).

6.4 Matched vs. Mismatched Analysis

Table 4 shows flip rates broken down by data split. The *flip rate* for a given model and perturbation method is defined as the proportion of examples that the model classified correctly on the original premise but incorrectly on the perturbed premise:

$$\text{flip rate} = \frac{|\{i : \text{correct}(\text{orig}_i) \wedge \neg\text{correct}(\text{pert}_i)\}|}{|\{i : \text{correct}(\text{orig}_i)\}|} \quad (1)$$

A higher flip rate indicates that the model is more sensitive to the syntactic perturbation.

Model	Perturbation	Match	Mismat	Over.
RoBERTa	passive	0.0925	0.1007	0.0966
BART	passive	0.0894	0.0979	0.0937
RoBERTa	adjrel	0.0217	0.0263	0.0240
BART	adjrel	0.0102	0.0131	0.0116
RoBERTa	pp	0.0126	0.0109	0.0117
BART	pp	0.0170	0.0098	0.0133
MLP	passive	0.0481	0.0466	0.0473
RNN	passive	0.0740	0.0764	0.0753

Table 4: Flip rates by model, perturbation, and data split

For passive voice, both RoBERTa and BART show higher failure rates on mismatched (out-of-domain) text than on matched (in-domain) text. This suggests that transformer models are more vulnerable to syntactic perturbations when dealing with out-of-domain text, because they cannot rely on domain-specific patterns learned during fine-tuning. Surprisingly, for prepositional phrase insertion, BART shows the opposite trend i.e. a lower failure rate on mismatched text suggesting that out-of-domain examples may contain simpler premise-hypothesis relationships that are easier to classify even after perturbation.

MLP and RNN show no consistent matched/mismatched pattern, with failure rates remaining relatively stable across splits, making them equally uncertain regardless of domain.

7 Complexity Metrics vs. Model Error Rate

Figure 1 shows scatter plots of average complexity metric values (X-axis) against model error rate (Y-axis) for each perturbation condition. Each point represents a (model, perturbation method) pair.

Two distinct clusters are visible across all three plots. The upper cluster (error rate ∼0.42–0.45) contains the MLP and RNN models, which maintain consistently high error rates regardless of perturbation type or complexity level. The lower

cluster (error rate ∼0.10) contains RoBERTa and BART, which are much more accurate.

Within the transformer cluster, passive voice consistently produces the highest error rate (∼0.17) across all three complexity metrics, confirming that it is the most disruptive perturbation. This is visible as an upward spike for RoBERTa and BART in all three plots. The remaining perturbations (adjrel, pp, and original) cluster near 0.10, suggesting that these they do not meaningfully increase error rates.

For the MLP and RNN, error rates remain flat across all perturbation conditions and complexity levels, indicating that these models are not sensitive to the specific type of syntactic perturbation applied.

Limitation: Since each model has only four data points (one per perturbation condition), it is not possible to draw conclusions about a continuous linear relationship between complexity and error rate. The plots are best interpreted as a comparison across discrete perturbations.

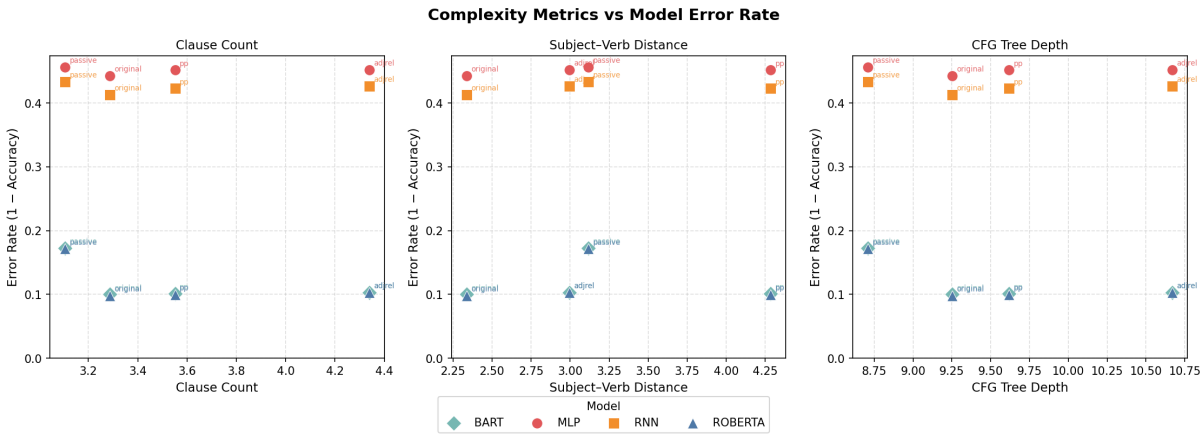

Figure 1: Scatter plots showing the relationship between average complexity metrics (X-axis) and model error rate (Y-axis) across perturbation conditions. Each point represents a (model, perturbation method) pair. Passive voice perturbation consistently produces the highest error rate for transformer models across all three complexity metrics.

References

Ronald T Kellogg. 1990. Effectiveness of prewriting strategies as a function of task demands. *The American Journal of Psychology*, pages 327–342.

Mohammadreza Valizadeh and Fatemeh Soltanpour. 2021. Focused direct corrective feedback: Effects on the elementary english learners' written syntactic complexity. *Eurasian Journal of Applied Linguistics*, 7(1):132–150.

Appendix

A List of Prepositional Phrases

The following phrases were used as the pool of generic prepositional phrases for the prepositional phrase insertion. Each phrase was selected to be semantically empty — carrying no referential or locational content that could introduce new information or contradict any hypothesis in the MultiNLI dataset.

- *at the time*
- *at that moment*
- *in question*
- *in this case*
- *at that point*
- *on that occasion*
- *in the scenario*
- *at that instance*

A phrase was selected uniformly at random (with seed 67) from this pool for each example.

B Error Summary

Table 5 shows the full error summary across all models, perturbation methods, and data splits, computed on a random sample of 2,000 examples.

Model	Perturbation	Split	Orig. Correct	Failures	Failure Rate
BART	adjrel	matched	884	9	0.0102
BART	adjrel	mismatched	919	12	0.0131
BART	adjrel	overall	1803	21	0.0116
BART	passive	matched	884	79	0.0894
BART	passive	mismatched	919	90	0.0979
BART	passive	overall	1803	169	0.0937
BART	pp	matched	884	15	0.0170
BART	pp	mismatched	919	9	0.0098
BART	pp	overall	1803	24	0.0133
MLP	adjrel	matched	520	29	0.0558
MLP	adjrel	mismatched	579	20	0.0345
MLP	adjrel	overall	1099	49	0.0446
MLP	passive	matched	520	25	0.0481
MLP	passive	mismatched	579	27	0.0466
MLP	passive	overall	1099	52	0.0473
MLP	pp	matched	520	27	0.0519
MLP	pp	mismatched	579	32	0.0553
MLP	pp	overall	1099	59	0.0537
RNN	adjrel	matched	554	40	0.0722
RNN	adjrel	mismatched	628	50	0.0796
RNN	adjrel	overall	1182	90	0.0761
RNN	passive	matched	554	41	0.0740
RNN	passive	mismatched	628	48	0.0764
RNN	passive	overall	1182	89	0.0753
RNN	pp	matched	554	38	0.0686
RNN	pp	mismatched	628	47	0.0748
RNN	pp	overall	1182	85	0.0719
RoBERTa	adjrel	matched	876	19	0.0217
RoBERTa	adjrel	mismatched	914	24	0.0263
RoBERTa	adjrel	overall	1790	43	0.0240
RoBERTa	passive	matched	876	81	0.0925
RoBERTa	passive	mismatched	914	92	0.1007
RoBERTa	passive	overall	1790	173	0.0966
RoBERTa	pp	matched	876	11	0.0126
RoBERTa	pp	mismatched	914	10	0.0109
RoBERTa	pp	overall	1790	21	0.0117

Table 5: Error summary across all models, perturbation methods, and data splits.